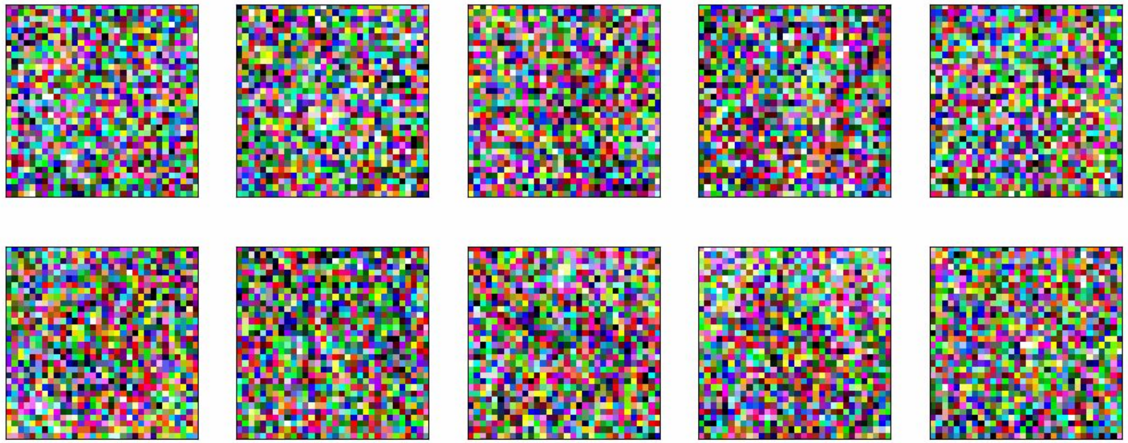
Guided Project: Latent Denoising Diffusion Probabilistic Models

Carnegie Mellon University

Introduction to Deep Learning Spring 2026

Preamble

- Story of image generation?
    - Diffusion is currently the best way
- Diffusion as the most powerful generative model
- Applications in other domains - Robot control



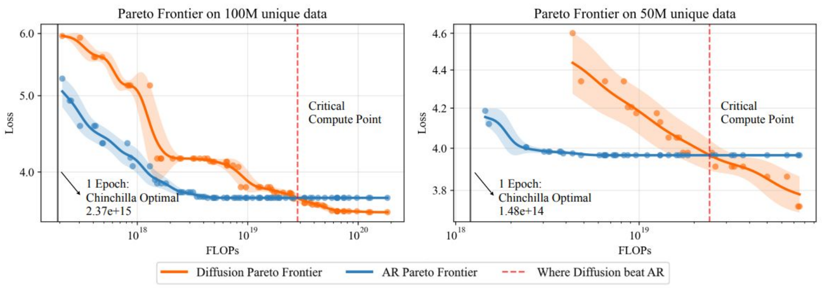
arXiv:2507.15857v6 [cs.LG] 15 Aug 2025

Diffusion Beats Autoregressive in Data-Constrained Settings

Mihir Prabhudesai* Carnegie Mellon University
Mengning Wu* Carnegie Mellon University
Amir Zadeh Lambda
Katerina Fragkiadaki Carnegie Mellon University
Deepak Pathak Carnegie Mellon University

Abstract

Autoregressive (AR) models have long dominated the landscape of large language models, driving progress across a wide range of tasks. Recently, diffusion-based language models have emerged as a promising alternative, though their advantages over AR models remain underexplored. In this paper, we systematically study masked diffusion models in data-constrained settings—where training involves repeated passes over limited data—and find that they significantly outperform AR models when compute is abundant but data is scarce. Diffusion models make better use of repeated data, achieving lower validation loss and superior downstream performance. We find new scaling laws for diffusion models and derive a closed-form expression for the critical compute threshold at which diffusion begins to outperform AR. Finally, we explain why diffusion models excel in this regime: their randomized masking objective implicitly trains over a rich distribution of token orderings, acting as an implicit data augmentation that AR's fixed left-to-right factorization lacks. Our results suggest that when data, not compute, is the bottleneck, diffusion models offer a compelling alternative to the standard AR paradigm. Our code is available at: `https://diffusion-scaling.github.io`.



)9.20328v2 [cs.LG] 29 Sep 2025

Google DeepMind 2025-10-1

Video models are zero-shot learners and reasoners

Thaddäus Wiedemer*[1], Yuxuan Li[1], Paul Vicol[1], Shixiang Shane Gu[1], Nick Matarese[1], Kevin Swersky[1], Been Kim[1], Priyank Jaini*[1] and Robert Geirhos*[1]
[1]Google DeepMind

The remarkable zero-shot capabilities of Large Language Models (LLMs) have propelled natural language processing from task-specific models to unified, generalist foundation models. This transformation emerged from simple primitives: large, generative models trained on web-scale data. Curiously, the same primitives apply to today's generative video models. Could video models be on a trajectory towards general-purpose *vision* understanding, much like LLMs developed general-purpose *language* understanding? We demonstrate that Veo 3 can solve a broad variety of tasks it wasn't explicitly trained for: segmenting objects, detecting edges, editing images, understanding physical properties, recognizing object affordances, simulating tool use, and more. These abilities to perceive, model, and manipulate the visual world enable early forms of visual reasoning like maze and symmetry solving. Veo's emergent zero-shot capabilities indicate that video models are on a path to becoming unified, generalist vision foundation models.
Project page: `https://video-zero-shot.github.io/`

1. Introduction

We believe that video models will become unifying, general-purpose foundation models for machine vision just like large language models (LLMs) have become foundation models for natural language processing (NLP). Within the last few years, NLP underwent a radical transformation: from task-specific, bespoke models (e.g., one model for translation, another one for question-answering, yet another one for summarization) to LLMs as unified foundation models. Today's LLMs are capable of general-purpose language understanding, which enables a single model to tackle a wide variety of tasks including coding [1], math [2], creative writing [3], summarization, translation [4], and deep research [5, 6]. These abilities started to emerge from simple primitives: training large, generative models on web-scale datasets [e.g 7, 8]. As a result, LLMs are increasingly able to solve novel tasks through few-shot in-context learning [7, 9] and zero-shot learning [10]. Zero-shot learning here means that prompting a model with a task instruction replaces the need for fine-tuning or adding task-specific inference heads.

Carnegie Mellon University

Homework outline

- Image diffusion model
- Implement a basic DDPM model
- explore advanced diffusion model training

Homework Outline - Necessary Implementations

- DDPM implementation from scratch.
- A DDIM noise scheduler that accelerates the generating process
- A Variational Autoencoder that maps the input space to the latent space for faster
- Following Classifier-Free Guidance during training and inference
- Evaluation of diffusion models.

Dataset - ImageNet 100

- 128 x 128 pixels images
- 100 Classes
- Approximately 130,000 images

"n01775062" : string "wolf spider, hunting spider"
"n01728572" : string "thunder snake, worm snake, Carphophis amoenus"
"n01601694" : string "water ouzel, dipper"
"n01978287" : string "Dungeness crab, Cancer magister"
"n01930112" : string "nematode, nematode worm, roundworm"
"n01739381" : string "vine snake"
"n01883070" : string "wombat"
"n01774384" : string "black widow, Latrodectus mactans"
"n02037110" : string "oystercatcher, oyster catcher"
"n01795545" : string "black grouse"
"n02027492" : string "red-backed sandpiper, dunlin, Erolia alpina"
"n01531178" : string "goldfinch, Carduelis carduelis"
"n01944390" : string "snail"
"n01494475" : string "hammerhead, hammerhead shark"
"n01632458" : string "spotted salamander, Ambystoma maculatum"
"n01698640" : string "American alligator, Alligator mississipiensis"
"n01675722" : string "banded gecko"
"n01877812" : string "wallaby, brush kangaroo"
"n01622779" : string "great grey owl, great gray owl, Strix nebulosa"
"n01910747" : string "jellyfish"
"n01860187" : string "black swan, Cygnus atratus"
"n01796340" : string "ptarmigan"

Goal: Model the Data Distribution

- Assume data samples x come from an unknown distribution

$$x \sim p_{\text{data}}(x)$$

- We want to model the data distribution

$$p_\theta(x) \;\approx\; p_{\text{data}}(x)$$

- Training Objective

$$\theta^* = \arg\max_\theta \; \mathbb{E}_{x \sim p_{\text{data}}}\Big[\log p_\theta(x)\Big]$$

VAE

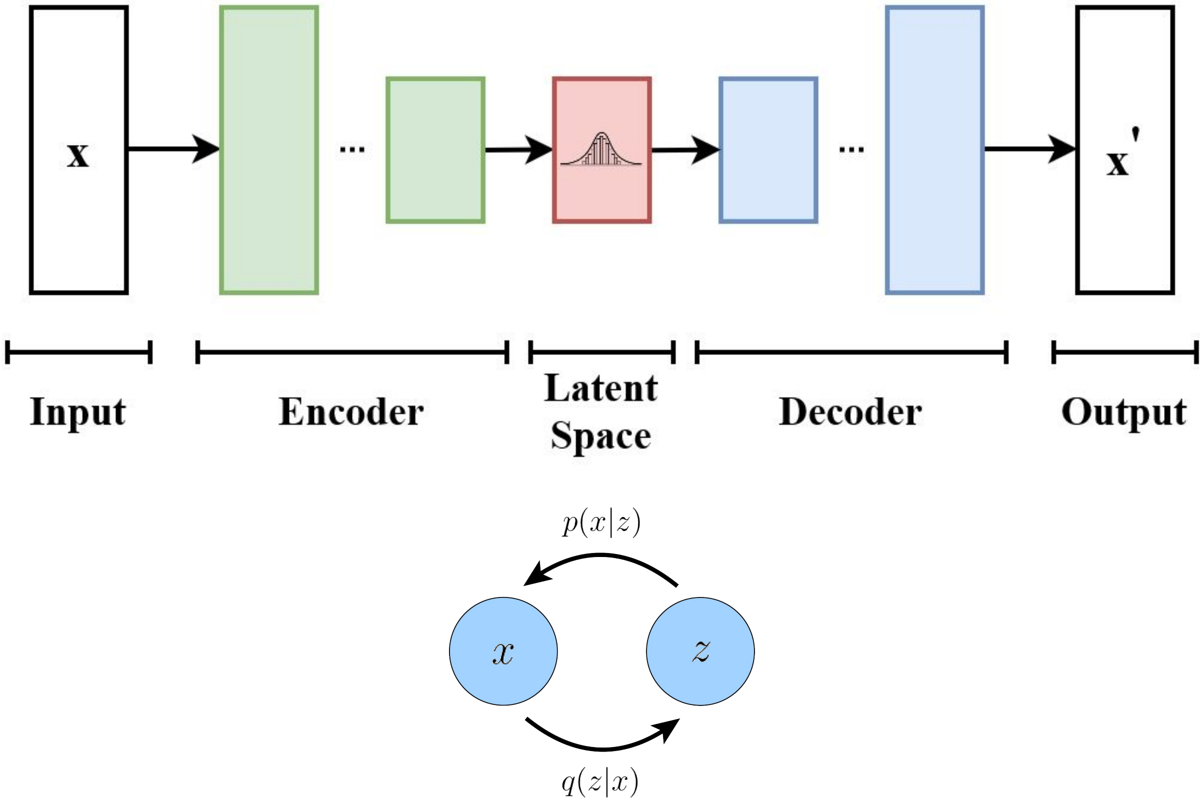
You will see it in lecture later...... 🥴

Auto-Encoding Variational Bayes

Diederik P. Kingma
Machine Learning Group
Universiteit van Amsterdam
dpkingma@gmail.com

Max Welling
Machine Learning Group
Universiteit van Amsterdam
welling.max@gmail.com

VAE: Architecture



VAE: Training Objective

- Latent Variable

$$p_\theta(x) = \int p_\theta(x, z)\, dz = \int p_\theta(x \mid z)\, p(z)\, dz$$

- Variational Lower Bound (ELBO)

$$\log p_\theta(x) \quad \geq \quad \mathbb{E}_{q_\phi(z|x)}\big[\log p_\theta(x \mid z)\big] - \mathrm{KL}\big(q_\phi(z \mid x) \,\|\, p(z)\big)$$

VAE: Training Objective

$$
\begin{aligned}
\ell(\theta; x) &= \ln p_\theta(x) \\
&= \ln \int_z p_\theta(x, z) && \text{Intractable} \\
&= \ln \int_z q_\phi(z|x) \frac{p_\theta(x, z)}{q_\phi(z)} \\
&= \ln \mathbb{E}_{q_\phi(z|x)} \left[\frac{p_\theta(x, z)}{q_\phi(z|x)} \right] \\
&\geq \mathbb{E}_{q_\phi(z|x)} \ln \left[\frac{p_\theta(x, z)}{q_\phi(z|x)} \right] && \text{Jensen's Inequality} \\
&= \int_z q_\phi(z|x) \ln \left[\frac{p_\theta(x, z)}{q_\phi(z|x)} \right] \\
&= \left[\int_z q_\phi(z|x) \ln p_\theta(x, z) - \int_z q_\phi(z|x) \ln q_\phi(z|x) \right] \\
&\triangleq \mathcal{L}(q, \theta)
\end{aligned}
$$

VAE: Training Objective

$$
\begin{aligned}
\mathcal{L}(q,\theta) &= \mathbb{E}_{z\sim q_\phi}[\ln p_\theta(x,z)] - \mathbb{E}_{z\sim q_\phi}[\ln q_\phi(z|x))] \\
&= \mathbb{E}_{z\sim q_\phi}[\ln p(z) + \ln p_\theta(x|z)] - \mathbb{E}_{z\sim q_\phi}[\ln q_\phi(z|x))] \\
&= \mathbb{E}_{z\sim q_\phi}[\ln p(z)] + \mathbb{E}_{z\sim q_\phi}[\ln p_\theta(x|z)] - \mathbb{E}_{z\sim q_\phi}[\ln q_\phi(z|x))] \\
&= \mathbb{E}_{z\sim q_\phi}[\ln p_\theta(x|z)] + \mathbb{E}_{z\sim q_\phi}[\ln p(z)] - \mathbb{E}_{z\sim q_\phi}[\ln q_\phi(z|x))] \\
&= \mathbb{E}_{z\sim q_\phi}[\ln p_\theta(x|z)] - KL(q_\phi(z|x)||p(z))
\end{aligned}
$$

Reconstruction Term Prior Matching Term

VAE: Training Objective

- Latent Variable

$$p_\theta(x) = \int p_\theta(x, z)\, dz = \int p_\theta(x \mid z)\, p(z)\, dz$$

- Variational Lower bound (ELBO)

$$\log p_\theta(x) \quad \geq \quad \mathbb{E}_{q_\phi(z|x)}\big[\log p_\theta(x \mid z)\big] - \mathrm{KL}\big(q_\phi(z \mid x) \,\|\, p(z)\big)$$

VAE: The Reparameterization Trick

- Use Gaussian posterior and likelihood distributions.
- Problem: Gradient cannot go through sampling
- Solution: the reparameterization trick

$$
\begin{aligned}
z &\sim N(\mu_z, \Sigma_z) \\
z &= \mu_z + \sqrt{\Sigma_z} \odot \epsilon \\
&= \mu_\phi(x) + \sqrt{\Sigma_\phi(x)} \odot \epsilon \quad , \epsilon \sim \mathcal{N}(0, I)
\end{aligned}
$$

VAE: Training Objective

- Approximate the reconstruction term
- The KL term can be solved analytically

VAE: Training Objective

$$p_\theta(x \mid z) = \mathcal{N}\big(x;\, \mu_\theta(z),\, \sigma_x^2 I\big)$$

$$\log p_\theta(x \mid z) = -\frac{D}{2}\log(2\pi\sigma_x^2) - \frac{1}{2\sigma_x^2}\,\|x - \mu_\theta(z)\|_2^2$$

$$\mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x \mid z)] = -\frac{D}{2}\log(2\pi\sigma_x^2) - \frac{1}{2\sigma_x^2}\,\mathbb{E}_{q_\phi(z|x)}[\|x - \mu_\theta(z)\|_2^2]$$

$$z = \mu_\phi(x) + \sigma_\phi(x) \odot \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I)$$

$$q_\phi(z \mid x) = \mathcal{N}\big(z;\, \mu_\phi(x),\, \mathrm{diag}(\sigma_\phi^2(x))\big)$$

$$p(z) = \mathcal{N}(0, I)$$

$$\mathrm{KL}(q_\phi(z \mid x) \,\|\, p(z)) = \frac{1}{2}\sum_{i=1}^{d}\Big(\mu_i^2 + \sigma_i^2 - \log\sigma_i^2 - 1\Big)$$

$$\mathcal{L}_{\mathrm{ELBO}}(x;\theta,\phi) = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x \mid z)] - \mathrm{KL}(q_\phi(z \mid x) \,\|\, p(z))$$

$$\mathcal{J}(x;\theta,\phi) = -\mathcal{L}_{\mathrm{ELBO}}(x;\theta,\phi) = \frac{1}{2\sigma_x^2}\|x - \mu_\theta(z)\|_2^2 + \frac{1}{2}\sum_{i}\Big(\mu_i^2 + \sigma_i^2 - \log\sigma_i^2 - 1\Big)$$

DDPM

You might see it in lecture later…… 🤔

Denoising Diffusion Probabilistic Models

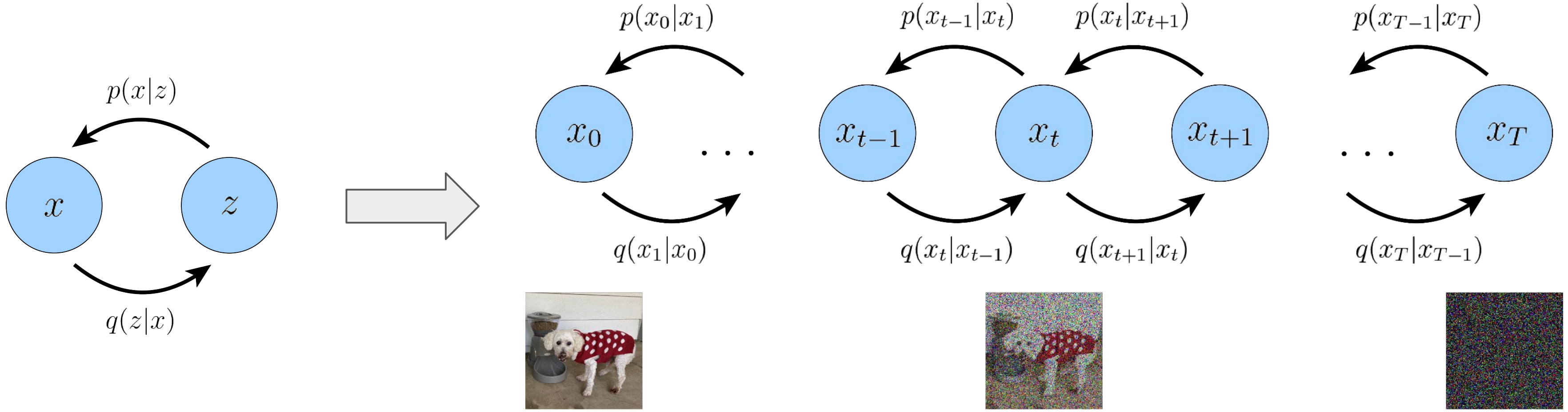
Jonathan Ho
UC Berkeley
jonathanho@berkeley.edu

Ajay Jain
UC Berkeley
ajayj@berkeley.edu

Pieter Abbeel
UC Berkeley
pabbeel@cs.berkeley.edu

DDPM: Architecture

- Stacking Many VAEs into a Markov Chain



DDPM: Forward Process

- Adding Noise

$$q(x_t \mid x_{t-1}) = \mathcal{N}(x_t; \sqrt{1-\beta_t}\, x_{t-1}, \beta_t I), \quad t = 1, \ldots, T.$$

$$q(x_{1:T} \mid x_0) = \prod_{t=1}^{T} q(x_t \mid x_{t-1}) \qquad 0 < \beta_1 < \beta_2 < \ldots \beta_T < 1$$

DDPM: Forward Process

$$\begin{aligned} x_t &= \sqrt{\alpha_t}\, x_{t-1} + \mathcal{N}(0, (1-\alpha_t)I) \\ &= \sqrt{\alpha_t}\, x_{t-1} + \sqrt{1-\alpha_t}\,\epsilon \quad , \epsilon \sim \mathcal{N}(0, I) \end{aligned}$$

$$\alpha_t = 1 - \beta_t$$

$$\begin{aligned} x_t &= \sqrt{\alpha_t}\; x_{t-1} + \sqrt{1-\alpha_t}\; \epsilon_t \\ &= \sqrt{\alpha_t}\left(\sqrt{\alpha_{t-1}}\; x_{t-2} + \sqrt{1-\alpha_{t-1}}\; \epsilon_{t-1}\right) + \sqrt{1-\alpha_t}\; \epsilon_t \\ &= \sqrt{\alpha_t\alpha_{t-1}}\; x_{t-2} + \sqrt{\alpha_t - \alpha_t\alpha_{t-1}}\; \epsilon_{t-1} + \sqrt{1-\alpha_t}\; \epsilon_t \\ &= \sqrt{\alpha_t\alpha_{t-1}}\; x_{t-2} + \sqrt{\sqrt{\alpha_t - \alpha_t\alpha_{t-1}}^2 + \sqrt{1-\alpha_t}^2}\; \epsilon \\ &= \sqrt{\alpha_t\alpha_{t-1}}\; x_{t-2} + \sqrt{1 - \alpha_t\alpha_{t-1}}\; \epsilon \\ &= \ldots \\ &= \sqrt{\prod_{i=1}^{t} \alpha_i}\; x_0 + \sqrt{1 - \prod_{i=1}^{t} \alpha_i}\; \epsilon \\ &= \sqrt{\bar{\alpha}_t}\; x_0 + \sqrt{1-\bar{\alpha}_t}\; \epsilon \quad , \bar{\alpha} = \prod_{i=1}^{t} \alpha_i, \quad \epsilon \sim \mathcal{N}(0, I) \\ &\sim \mathcal{N}(\sqrt{\bar{\alpha}_t}\; x_0, (1-\bar{\alpha}_t)I) \end{aligned}$$

DDPM: Forward Process

- Closed Form

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1 - \bar{\alpha}_t}\, \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I)$$

DDPM: Reverse Process

- Denoising, need to model it

$$q(x_t \mid x_{t-1}) = \mathcal{N}(x_t; \sqrt{1-\beta_t}\, x_{t-1}, \beta_t I)$$

$$p_\theta(x_{t-1} \mid x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t))$$

$$p_\theta(x_{0:T}) = p(x_T) \prod_{t=1}^{T} p_\theta(x_{t-1} \mid x_t)$$

DDPM: Training Objective

- Data Likelihood:

$$p_\theta(x_0) = \int \frac{p_\theta(x_{0:T})}{q(x_{1:T} \mid x_0)} q(x_{1:T} \mid x_0)\, dx_{1:T}$$

DDPM: Training Objective

$$
\begin{aligned}
\ln p(x_0) &\geq \mathbb{E}_{q(x_{1:T}|x_0)}\left[\ln \frac{p(x_{0:T})}{q(x_{1:T}|x_0)}\right] \\
&= \mathbb{E}_{q(x_{1:T}|x_0)}\left[\ln \frac{p(x_T)\prod_{t=1}^{T} p_\theta(x_{t-1}|x_t)}{\prod_{t=1}^{T} q(x_t|x_{t-1})}\right] \\
&= \mathbb{E}_{q(x_{1:T}|x_0)}\left[\ln \frac{p(x_T)p_\theta(x_0|x_1)\prod_{t=2}^{T} p_\theta(x_{t-1}|x_t)}{q(x_1|x_0)\prod_{t=2}^{T} q(x_t|x_{t-1})}\right] \\
&= \mathbb{E}_{q(x_{1:T}|x_0)}\left[\ln \frac{p(x_T)p_\theta(x_0|x_1)\prod_{t=2}^{T} p_\theta(x_{t-1}|x_t)}{q(x_1|x_0)\prod_{t=2}^{T} q(x_t|x_{t-1}, x_0)}\right] \\
&= \mathbb{E}_{q(x_{1:T}|x_0)}\left[\ln \frac{p_\theta(x_T)p_\theta(x_0|x_1)}{q(x_1|x_0)} + \ln \prod_{t=2}^{T} \frac{p_\theta(x_{t-1}|x_t)}{q(x_t|x_{t-1}, x_0)}\right] \\
&= \mathbb{E}_{q(x_{1:T}|x_0)}\left[\ln \frac{p(x_T)p_\theta(x_0|x_1)}{q(x_1|x_0)} + \ln \prod_{t=2}^{T} \frac{p_\theta(x_{t-1}|x_t)}{\frac{p(x_{t-1}|x_t,x_0)q(x_t|x_0)}{q(x_{t-1}|x_0)}}\right]
\end{aligned}
$$

$$
\begin{aligned}
&= \mathbb{E}_{q(x_{1:T}|x_0)}\left[\ln \frac{p(x_T)p_\theta(x_0|x_1)}{q(x_1|x_0)} + \ln \prod_{t=2}^{T} \frac{p_\theta(x_{t-1}|x_t)}{\frac{p(x_{t-1}|x_t,x_0)\,\cancel{q(x_t|x_0)}}{\cancel{q(x_{t-1}|x_0)}}}\right] \\
&= \mathbb{E}_{q(x_{1:T}|x_0)}\left[\ln \frac{p(x_T)p_\theta(x_0|x_1)}{\cancel{q(x_1|x_0)}} + \ln \frac{\cancel{q(x_1|x_0)}}{q(x_T|x_0)} + \ln \prod_{t=2}^{T} \frac{p_\theta(x_{t-1}|x_t)}{q(x_{t-1}|x_t, x_0)}\right] \\
&= \mathbb{E}_{q(x_{1:T}|x_0)}\left[\ln \frac{p(x_T)p_\theta(x_0|x_1)}{q(x_T|x_0)} + \sum_{t=2}^{T} \ln \frac{p_\theta(x_{t-1}|x_t)}{q(x_{t-1}|x_t, x_0)}\right] \\
&= \mathbb{E}_{q(x_{1:T}|x_0)}\left[\ln p_\theta(x_0|x_1)\right] + \mathbb{E}_{q(x_{1:T}|x_0)}\left[\ln \frac{p(x_T)}{q(x_T|x_0)}\right] + \sum_{t=2}^{T} \mathbb{E}_{q(x_{1:T}|x_0)}\left[\ln \frac{p_\theta(x_{t-1}|x_t)}{q(x_{t-1}|x_t, x_0)}\right] \\
&= \mathbb{E}_{q(x_1|x_0)}\left[\ln p_\theta(x_0|x_1)\right] + \mathbb{E}_{q(x_T|x_0)}\left[\ln \frac{p(x_T)}{q(x_T|x_0)}\right] + \sum_{t=2}^{T} \mathbb{E}_{q(x_t, x_{t-1}|x_0)}\left[\ln \frac{p_\theta(x_{t-1}|x_t)}{q(x_{t-1}|x_t, x_0)}\right] \\
&= \underbrace{\mathbb{E}_{q(x_1|x_0)}\left[\ln p_\theta(x_0|x_1)\right]}_{\text{reconstruction term}} - \underbrace{D_{\mathrm{KL}}\left(q(x_T|x_0) \,\|\, p(x_T)\right)}_{\text{prior matching term}} \\
&\quad - \sum_{t=2}^{T} \underbrace{\mathbb{E}_{q(x_t|x_0)}\left[D_{\mathrm{KL}}\left(q(x_{t-1}|x_t, x_0) \,\|\, p_\theta(x_{t-1}|x_t)\right)\right]}_{\text{denoising matching term}}
\end{aligned}
$$

DDPM: Training Objective

- Data Likelihood:

$$p_\theta(x_0) = \int \frac{p_\theta(x_{0:T})}{q(x_{1:T} \mid x_0)} q(x_{1:T} \mid x_0)\, dx_{1:T}$$

- Variational Lower Bound (ELBO):

$$\ln p(x_0) \geq \underbrace{\mathbb{E}_{q(x_1|x_0)}\left[\ln p_\theta(x_0|x_1)\right]}_{\text{reconstruction term}} - \underbrace{D_{\mathrm{KL}}\left(q(x_T|x_0) \| p(x_T)\right)}_{\text{prior matching term}}$$

$$- \sum_{t=2}^{T} \underbrace{\mathbb{E}_{q(x_t|x_0)}\left[D_{\mathrm{KL}}\left(q(x_{t-1}|x_t, x_0) \| p_\theta(x_{t-1}|x_t)\right)\right]}_{\text{denoising matching term}}$$

DDPM: Denoising Matching Term

- $$-\sum_{t=2}^{T} \underbrace{\mathbb{E}_{q(x_t|x_0)} \left[D_{\mathrm{KL}} \left(q(x_{t-1}|x_t, x_0) \| p_\theta(x_{t-1}|x_t) \right) \right]}_{\text{denoising matching term}}$$

DDPM: Denoising Matching Term

$$
\begin{aligned}
q(x_{t-1}|x_t,x_0) &= \frac{q(x_t|x_{t-1},x_0)q(x_{t-1}|x_0)}{q(x_t|x_0)}\\
&= \frac{\mathcal{N}(x_t;\sqrt{\alpha_t}x_{t-1},(1-\alpha_t)I)\mathcal{N}(x_{t-1};\sqrt{\bar\alpha_{t-1}}x_0,(1-\bar\alpha_{t-1})I)}{\mathcal{N}(x_t;\sqrt{\bar\alpha_t}x_0,(1-\bar\alpha_t)I)}\\
&\propto \exp\left\{-\left[\frac{(x_t-\sqrt{\alpha_t}x_{t-1})^2}{2(1-\alpha_t)}+\frac{(x_{t-1}-\sqrt{\bar\alpha_{t-1}}x_0)^2}{2(1-\bar\alpha_{t-1})}-\frac{(x_t-\sqrt{\bar\alpha_t}x_0)^2}{2(1-\bar\alpha_t)}\right]\right\}\\
&= \exp\left\{-\frac{1}{2}\left[\frac{(x_t-\sqrt{\alpha_t}x_{t-1})^2}{1-\alpha_t}+\frac{(x_{t-1}-\sqrt{\bar\alpha_{t-1}}x_0)^2}{1-\bar\alpha_{t-1}}-\frac{(x_t-\sqrt{\bar\alpha_t}x_0)^2}{1-\bar\alpha_t}\right]\right\}\\
&= \exp\left\{-\frac{1}{2}\left[\frac{(-2\sqrt{\alpha_t}x_tx_{t-1}+\alpha_tx_{t-1}^2)}{1-\alpha_t}+\frac{(x_{t-1}^2-2\sqrt{\bar\alpha_{t-1}}x_{t-1}x_0)}{1-\bar\alpha_{t-1}}+C(x_t,x_0)\right]\right\}\\
&\propto \exp\left\{-\frac{1}{2}\left[-\frac{2\sqrt{\alpha_t}x_tx_{t-1}}{1-\alpha_t}+\frac{\alpha_tx_{t-1}^2}{1-\alpha_t}+\frac{x_{t-1}^2}{1-\bar\alpha_{t-1}}-\frac{2\sqrt{\bar\alpha_{t-1}}x_{t-1}x_0}{1-\bar\alpha_{t-1}}\right]\right\}\\
&= \exp\left\{-\frac{1}{2}\left[(\frac{\alpha_t}{1-\alpha_t}+\frac{1}{1-\bar\alpha_{t-1}})x_{t-1}^2-2\left(\frac{\sqrt{\alpha_t}x_t}{1-\alpha_t}+\frac{\sqrt{\bar\alpha_{t-1}}x_0}{1-\bar\alpha_{t-1}}\right)x_{t-1}\right]\right\}\\
&= \exp\left\{-\frac{1}{2}\left[\frac{\alpha_t(1-\bar\alpha_{t-1})+1-\alpha_t}{(1-\alpha_t)(1-\bar\alpha_{t-1})}x_{t-1}^2-2\left(\frac{\sqrt{\alpha_t}x_t}{1-\alpha_t}+\frac{\sqrt{\bar\alpha_{t-1}}x_0}{1-\bar\alpha_{t-1}}\right)x_{t-1}\right]\right\}
\end{aligned}
$$

$$
\begin{aligned}
&= \exp\left\{-\frac{1}{2}\left[\frac{\alpha_t-\bar\alpha_t+1-\alpha_t}{(1-\alpha_t)(1-\bar\alpha_{t-1})}x_{t-1}^2-2\left(\frac{\sqrt{\alpha_t}x_t}{1-\alpha_t}+\frac{\sqrt{\bar\alpha_{t-1}}x_0}{1-\bar\alpha_{t-1}}\right)x_{t-1}\right]\right\}\\
&= \exp\left\{-\frac{1}{2}\left[\frac{1-\bar\alpha_t}{(1-\alpha_t)(1-\bar\alpha_{t-1})}x_{t-1}^2-2\left(\frac{\sqrt{\alpha_t}x_t}{1-\alpha_t}+\frac{\sqrt{\bar\alpha_{t-1}}x_0}{1-\bar\alpha_{t-1}}\right)x_{t-1}\right]\right\}\\
&= \exp\left\{-\frac{1}{2}\left(\frac{1-\bar\alpha_t}{(1-\alpha_t)(1-\bar\alpha_{t-1})}\right)\left[x_{t-1}^2-2\frac{\left(\frac{\sqrt{\alpha_t}x_t}{1-\alpha_t}+\frac{\sqrt{\bar\alpha_{t-1}}x_0}{1-\bar\alpha_{t-1}}\right)}{\frac{1-\bar\alpha_t}{(1-\alpha_t)(1-\bar\alpha_{t-1})}}x_{t-1}\right]\right\}\\
&= \exp\left\{-\frac{1}{2}\left(\frac{1-\bar\alpha_t}{(1-\alpha_t)(1-\bar\alpha_{t-1})}\right)\left[x_{t-1}^2-2\frac{\left(\frac{\sqrt{\alpha_t}x_t}{1-\alpha_t}+\frac{\sqrt{\bar\alpha_{t-1}}x_0}{1-\bar\alpha_{t-1}}\right)(1-\alpha_t)(1-\bar\alpha_{t-1})}{1-\bar\alpha_t}x_{t-1}\right]\right\}\\
&= \exp\left\{-\frac{1}{2}\left(\frac{1}{\frac{(1-\alpha_t)(1-\bar\alpha_{t-1})}{1-\bar\alpha_t}}\right)\left[x_{t-1}^2-2\frac{\sqrt{\alpha_t}(1-\bar\alpha_{t-1})x_t+\sqrt{\bar\alpha_{t-1}}(1-\alpha_t)x_0}{1-\bar\alpha_t}x_{t-1}\right]\right\}\\
&\propto \mathcal{N}(x_{t-1};\underbrace{\frac{\sqrt{\alpha_t}(1-\bar\alpha_{t-1})x_t+\sqrt{\bar\alpha_{t-1}}(1-\alpha_t)x_0}{1-\bar\alpha_t}}_{\mu_q(x_t,x_0)},\underbrace{\frac{(1-\alpha_t)(1-\bar\alpha_{t-1})}{1-\bar\alpha_t}I}_{\Sigma_q(t)})
\end{aligned}
$$

DDPM: Denoising Matching Term

$$
\begin{aligned}
& D_{\mathrm{KL}}\left(q(x_{t-1}|x_t, x_0) \| p_\theta(x_{t-1}|x_t)\right) \\
&= D_{\mathrm{KL}}\left(\mathcal{N}(x_{t-1}; \mu_q, \Sigma_q(t)) \| \mathcal{N}(x_{t-1}; \mu_\theta, \Sigma_q(t))\right) \\
&= \frac{1}{2}\left[\log \frac{|\Sigma_q(t)|}{|\Sigma_q(t)|} - d + \operatorname{tr}(\Sigma_q(t)^{-1}\Sigma_q(t)) + (\mu_\theta - \mu_q)^T \Sigma_q(t)^{-1}(\mu_\theta - \mu_q)\right] \\
&= \frac{1}{2}\left[\log 1 - d + d + (\mu_\theta - \mu_q)^T \Sigma_q(t)^{-1}(\mu_\theta - \mu_q)\right] \\
&= \frac{1}{2}\left[(\mu_\theta - \mu_q)^T \Sigma_q(t)^{-1}(\mu_\theta - \mu_q)\right] \\
&= \frac{1}{2}\left[(\mu_\theta - \mu_q)^T \left(\sigma_q^2(t)\mathbf{I}\right)^{-1}(\mu_\theta - \mu_q)\right] \\
&= \frac{1}{2\sigma_q^2(t)}\left[\|\mu_\theta - \mu_q\|_2^2\right]
\end{aligned}
$$

DDPM: Denoising Matching Term

- $$q(x_{t-1}|x_t, x_0) \propto \mathcal{N}(x_{t-1}; \underbrace{\frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})x_t + \sqrt{\bar{\alpha}_{t-1}}(1-\alpha_t)x_0}{1-\bar{\alpha}_t}}_{\mu_q(x_t, x_0)}, \underbrace{\frac{(1-\alpha_t)(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}I}_{\Sigma_q(t)})$$

- $$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \epsilon_t \quad , \bar{\alpha}_t = \prod_{i=1}^{t} \alpha_i, \;\; \epsilon_t \sim \mathcal{N}(0, I)$$

$$x_0 = \frac{x_t - \sqrt{1-\bar{\alpha}_t}\, \epsilon_t}{\sqrt{\bar{\alpha}_t}}, \;\; \epsilon_t \sim \mathcal{N}(0, I)$$

DDPM: Denoising Matching Term

$$
\begin{aligned}
\mu_q = \mu_q(x_t, x_0) &= \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})x_t + \sqrt{\bar{\alpha}_{t-1}}(1-\alpha_t)x_0}{1-\bar{\alpha}_t} \\
&= \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})x_t + \sqrt{\bar{\alpha}_{t-1}}(1-\alpha_t)\frac{x_t-\sqrt{1-\bar{\alpha}_t}\epsilon}{\sqrt{\bar{\alpha}_t}}}{1-\bar{\alpha}_t} \\
&= \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})x_t + (1-\alpha_t)\frac{x_t-\sqrt{1-\bar{\alpha}_t}\epsilon}{\sqrt{\alpha_t}}}{1-\bar{\alpha}_t} \\
&= \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})x_t}{1-\bar{\alpha}_t} + \frac{(1-\alpha_t)x_t}{(1-\bar{\alpha}_t)\sqrt{\alpha_t}} - \frac{(1-\alpha_t)\sqrt{1-\bar{\alpha}_t}\epsilon}{(1-\bar{\alpha}_t)\sqrt{\alpha_t}} \\
&= \left(\frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t} + \frac{1-\alpha_t}{(1-\bar{\alpha}_t)\sqrt{\alpha_t}}\right)x_t - \frac{(1-\alpha_t)\sqrt{1-\bar{\alpha}_t}}{(1-\bar{\alpha}_t)\sqrt{\alpha_t}}\epsilon \\
&= \left(\frac{\alpha_t(1-\bar{\alpha}_{t-1})}{(1-\bar{\alpha}_t)\sqrt{\alpha_t}} + \frac{1-\alpha_t}{(1-\bar{\alpha}_t)\sqrt{\alpha_t}}\right)x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}\sqrt{\alpha_t}}\epsilon \\
&= \frac{\alpha_t-\bar{\alpha}_t+1-\alpha_t}{(1-\bar{\alpha}_t)\sqrt{\alpha_t}}x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}\sqrt{\alpha_t}}\epsilon \\
&= \frac{1-\bar{\alpha}_t}{(1-\bar{\alpha}_t)\sqrt{\alpha_t}}x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}\sqrt{\alpha_t}}\epsilon \\
&= \frac{1}{\sqrt{\alpha_t}}x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}\sqrt{\alpha_t}}\epsilon
\end{aligned}
$$

DDPM: Denoising Matching Term

- Instead of modeling the mean of the backward process at time t, we choose to model the noise at time t.

$$\mu_\theta = \mu_\theta(x_t, t) = \frac{1}{\sqrt{\alpha_t}} x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}\sqrt{\alpha_t}} \hat{\epsilon}_\theta(x_t, t)$$

$$\begin{aligned}
& D_{\mathrm{KL}}\left(q(x_{t-1}|x_t, x_0) \| p_\theta(x_{t-1}|x_t)\right) \\
&= \frac{1}{2\sigma_q^2(t)} \left[\|\mu_\theta - \mu_q\|_2^2\right] \\
&= \frac{1}{2\sigma_q^2(t)} \left[\left\|\frac{1}{\sqrt{\alpha_t}} x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}\sqrt{\alpha_t}} \hat{\epsilon}_\theta(x_t, t) - \frac{1}{\sqrt{\alpha_t}} x_t + \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}\sqrt{\alpha_t}} \epsilon\right\|_2^2\right] \\
&= \frac{1}{2\sigma_q^2(t)} \left[\left\|\frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}\sqrt{\alpha_t}} \epsilon - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}\sqrt{\alpha_t}} \hat{\epsilon}_\theta(x_t, t)\right\|_2^2\right] \\
&= \frac{1}{2\sigma_q^2(t)} \left[\left\|\frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}\sqrt{\alpha_t}} (\epsilon - \hat{\epsilon}_\theta(x_t, t))\right\|_2^2\right] \\
&= \frac{1}{2\sigma_q^2(t)} \frac{(1-\alpha_t)^2}{(1-\bar{\alpha}_t)\alpha_t} \left[\|\epsilon - \hat{\epsilon}_\theta(x_t, t)\|_2^2\right]
\end{aligned}$$

DDPM: Reconstruction Term

$$\underbrace{\mathbb{E}_{q(x_1|x_0)}\left[\ln p_\theta(x_0|x_1)\right]}_{\text{reconstruction term}}$$

- Define the last reverse process as:

$$p_\theta(\mathbf{x}_0|\mathbf{x}_1) = \prod_{i=1}^{D} \int_{\delta_-(x_0^i)}^{\delta_+(x_0^i)} \mathcal{N}(x; \mu_\theta^i(\mathbf{x}_1, 1), \sigma_1^2)\, dx$$

$$\delta_+(x) = \begin{cases} \infty & \text{if } x = 1 \\ x + \frac{1}{255} & \text{if } x < 1 \end{cases} \qquad \delta_-(x) = \begin{cases} -\infty & \text{if } x = -1 \\ x - \frac{1}{255} & \text{if } x > -1 \end{cases}$$

DDPM: Prior Matching Term

$$\underbrace{D_{\mathrm{KL}}\left(q(x_T|x_0)\|p(x_T)\right)}_{\text{prior matching term}}$$

- No Parameters! 🥳

DDPM: Training Objective

- We can simplify the ELBO based on our derivations

$$L_{\text{simple}}(\theta) := \mathbb{E}_{t,\mathbf{x}_0,\boldsymbol{\epsilon}}\left[\left\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, t)\right\|^2\right]$$

DDPM: Training and Inference

Algorithm 1 Training

1: **repeat**
2: $x_0 \sim q(x_0)$ ▷*Sample random initial data*
3: $t \sim \text{Uniform}(\{1, ..., T\})$ ▷*Sample random step*
4: $\epsilon \sim \mathcal{N}(0, I)$ ▷*Sample random noise*
5: $x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon$ ▷*Rand. step of rand. trajectory*
6: Take gradient descent step on $\nabla_\theta ||\epsilon - \epsilon_\theta(x_t, t)||^2$ ▷*Optimisation*
7: **until** converged

Algorithm 2 Sampling

1: $x_T \sim \mathcal{N}(0, I)$ ▷*Initial isotropic gaussian noise sampling*
2: **for** t = T, ..., 1 **do**
3: $z \sim \mathcal{N}(0, I)$ if $t > 1$ else $z = 0$ ▷*Sample random noise (if not last step)*
4: $\tilde{\epsilon} = \epsilon_\theta(x_t, t)$ ▷*Estimated noise in current noisy data*
5: $\tilde{x_0} = \frac{1}{\sqrt{\bar{\alpha}_t}}(x_t - \sqrt{1 - \bar{\alpha}_t}\tilde{\epsilon})$ ▷*Estimated x_0 from estimated noise*
6: $\tilde{\mu} = \mu_t(x_t, \tilde{x_0}) \left(= \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_\theta(x_t, t)\right)\right)$ ▷*Mean for previous step sampling*
7: $x_{t-1} = \tilde{\mu} + \sigma_t z$ ▷*Previous step sampling*
8: **end for**
9: **return** x_0

DDIM

- You might not see it in the lectures 😞

DENOISING DIFFUSION IMPLICIT MODELS

Jiaming Song, Chenlin Meng & Stefano Ermon
Stanford University
{tsong,chenlin,ermon}@cs.stanford.edu

DDIM: Motivation

- Recall DDPM Inference

Algorithm 2 Sampling

$x_T \sim \mathcal{N}(0, I)$ ▷*Initial isotropic gaussian noise sampling*
for t = T, ..., 1 **do**
  $z \sim \mathcal{N}(0, I)$ if $t > 1$ else $z = 0$ ▷*Sample random noise (if not last step)*
  $\tilde{\epsilon} = \epsilon_\theta(x_t, t)$ ▷*Estimated noise in current noisy data*
  $\tilde{x_0} = \frac{1}{\sqrt{\bar{\alpha}_t}}(x_t - \sqrt{1-\bar{\alpha}_t}\tilde{\epsilon})$ ▷*Estimated x_0 from estimated noise*
  $\tilde{\mu} = \mu_t(x_t, \tilde{x_0}) \left(= \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_\theta(x_t, t)\right)\right)$ ▷*Mean for previous step sampling*
  $x_{t-1} = \tilde{\mu} + \sigma_t z$ ▷*Previous step sampling*
end for
return x_0

- Can we skip some steps

DDIM: Motivation

- DDPM Training Objective

$$L_{\text{simple}}(\theta) := \mathbb{E}_{t,\mathbf{x}_0,\boldsymbol{\epsilon}}\left[\left\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, t)\right\|^2\right]$$

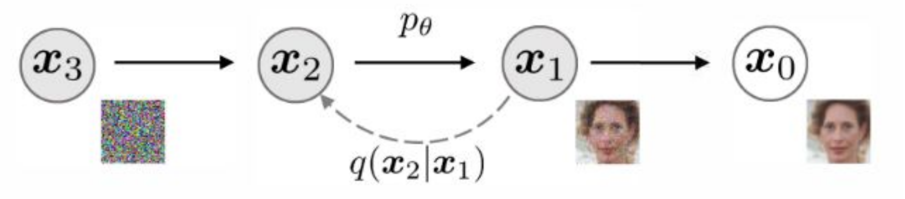
$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \epsilon_t \quad , \bar{\alpha}_t = \prod_{i=1}^{t} \alpha_i, \;\; \epsilon_t \sim \mathcal{N}(0, I)$$

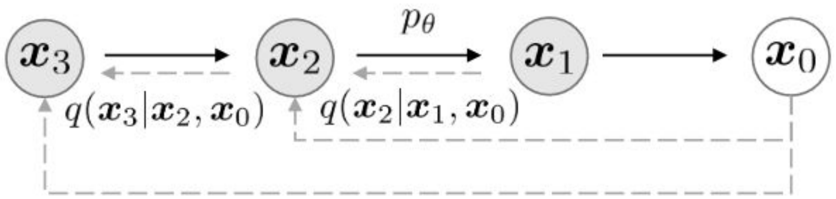
- Only depends on $q(x_t|x_0)$

DDIM: Motivation

- Redesign forward pass while keeping the marginals unchanged

$$q_\sigma(x_t|x_0) \sim \mathcal{N}(\sqrt{\bar{\alpha}_t}\, x_0, (1-\bar{\alpha}_t)I)$$





DDIM: Forward Process

- Define the following new distributions

$$q_\sigma(\boldsymbol{x}_{1:T}|\boldsymbol{x}_0) := q_\sigma(\boldsymbol{x}_T|\boldsymbol{x}_0)\prod_{t=2}^{T} q_\sigma(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t, \boldsymbol{x}_0)$$

$$q_\sigma(\boldsymbol{x}_T|\boldsymbol{x}_0) = \mathcal{N}(\sqrt{\alpha_T}\boldsymbol{x}_0, (1-\alpha_T)\boldsymbol{I})$$

$$q_\sigma(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t, \boldsymbol{x}_0) = \mathcal{N}\left(\sqrt{\alpha_{t-1}}\boldsymbol{x}_0 + \sqrt{1-\alpha_{t-1}-\sigma_t^2}\cdot\frac{\boldsymbol{x}_t - \sqrt{\alpha_t}\boldsymbol{x}_0}{\sqrt{1-\alpha_t}}, \sigma_t^2\boldsymbol{I}\right)$$

- Can be proved that the marginals still hold

$$q_\sigma(x_t|x_0) \sim \mathcal{N}(\sqrt{\bar{\alpha}_t}\, x_0, (1-\bar{\alpha}_t)I)$$

DDPM: Training Objective

- Variational Lower Bound (ELBO)

$$J_\sigma(\epsilon_\theta) := \mathbb{E}_{\boldsymbol{x}_{0:T} \sim q_\sigma(\boldsymbol{x}_{0:T})}[\log q_\sigma(\boldsymbol{x}_{1:T}|\boldsymbol{x}_0) - \log p_\theta(\boldsymbol{x}_{0:T})]$$
$$= \mathbb{E}_{\boldsymbol{x}_{0:T} \sim q_\sigma(\boldsymbol{x}_{0:T})}\left[\log q_\sigma(\boldsymbol{x}_T|\boldsymbol{x}_0) + \sum_{t=2}^{T} \log q_\sigma(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t, \boldsymbol{x}_0) - \sum_{t=1}^{T} \log p_\theta^{(t)}(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t) - \log p_\theta(\boldsymbol{x}_T)\right]$$

DDIM: Backward Process

- Define the following new distributions

$$p_\theta^{(t)}(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t) = \begin{cases} \mathcal{N}(f_\theta^{(1)}(\boldsymbol{x}_1), \sigma_1^2\boldsymbol{I}) & \text{if } t = 1 \\ q_\sigma(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t, f_\theta^{(t)}(\boldsymbol{x}_t)) & \text{otherwise,} \end{cases}$$

$$f_\theta^{(t)}(\boldsymbol{x}_t) := (\boldsymbol{x}_t - \sqrt{1-\alpha_t} \cdot \epsilon_\theta^{(t)}(\boldsymbol{x}_t))/\sqrt{\alpha_t}$$

- Can be proved that the new ELBO is equivalent to DDPM's ELBO up to a constant

DDIM: Backward Process

- Given the new definitions, do samping by

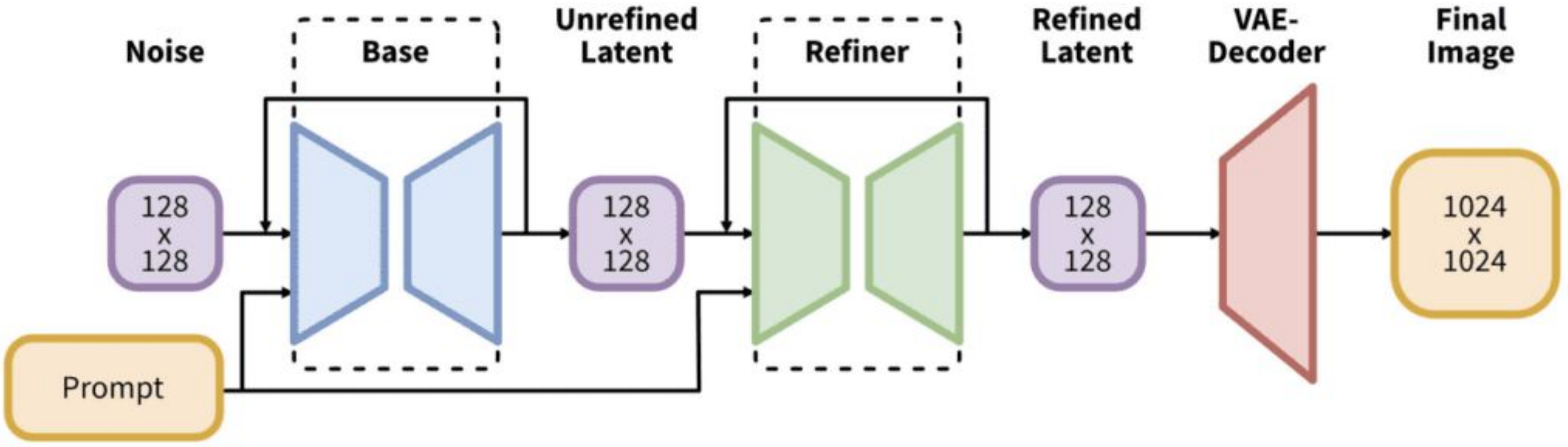
$$
\begin{aligned}
x_{t-1} &= \sqrt{\bar{\alpha}_{t-1}}\,\hat{x}_0 + \sqrt{1-\bar{\alpha}_{t-1}-\sigma_t^2}\cdot\frac{x_t-\sqrt{\bar{\alpha}_t}\,\hat{x}_0}{\sqrt{1-\bar{\alpha}_t}} + \sigma_t\epsilon_t^* \\
&= \underbrace{\sqrt{\bar{\alpha}_{t-1}}\left(\frac{x_t-\sqrt{1-\bar{\alpha}_t}\,\hat{\epsilon}_t(x_t,t)}{\sqrt{\bar{\alpha}_t}}\right)}_{\text{predict } x_0} + \underbrace{\sqrt{1-\bar{\alpha}_{t-1}-\sigma_t^2}\quad\hat{\epsilon}_t(x_t,t)}_{\text{direction pointing to } x_t} + \underbrace{\sigma_t\epsilon_t^*}_{\text{random noise}} \\
&\text{where}\quad \epsilon_t^* \sim \mathcal{N}(0,I)
\end{aligned}
$$

DDIM: Backward Process

- There's no requirement on the variances.
- If we set it to 0:
  - The sampling process is reproducible given initial noise.
  - Need fewer denoising steps.

Latent Diffusion Models (LDMs)

- Diffusion in the pixel-space is **expensive**
- The latent space captures data in a more compact and structured form
- We perform expensive diffusion in this space for **faster training**
- This is the key backbone behind stable diffusion (DALLE, ImageGen)



LDMs: Role of the VAE

- The Variational AutoEncoder is responsible for generating the latent space
- Its objective is to minimize reconstruction error while enforcing a distribution on the latent space

$$\mathcal{L}(\theta, \phi; \mathbf{x}) = \underbrace{\mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}[\log p_\theta(\mathbf{x}|\mathbf{z})]}_{\text{Reconstruction Loss}} - \underbrace{D_{\mathrm{KL}}(q_\phi(\mathbf{z}|\mathbf{x}) \| p(\mathbf{z}))}_{\text{KL Divergence Loss}}$$

👉 Your task is to implement the encoding part of the VAE that provides input to the DDPM and to transform the denoised output of the DDPM

Classifier Free Guidance

- Our current diffusion outputs aren't **explicitly structured**
- We would like to introduce some form of conditional generation
  - This can be on text, other images
- Previous methods would be classifier-based
  - Included a classification to adjust diffusion steps: **outdated**
- Nowadays, the model is trained conditionally and unconditionally which improves the quality and alignment of the generated sample

Classifier Free Guidance

- The model is trained with and without conditioning information
- During inference, CFG is implemented by interpolating between the conditional and unconditional score functions

$$\text{guided score} = \underbrace{\nabla_x \log p_\theta(x_t)}_{\text{realism}} + (1+w)\cdot \underbrace{(\nabla_x \log p_\theta(x_t \mid y) - \nabla_x \log p_\theta(x_t))}_{\text{condition contribution}}$$

👉 Your task is to setup the code for CFG and write the necessary documentation. You'll have to adjust the score functions for the reverse process

Evaluation of Diffusion Models

How do we evaluate the quality of a generated signal?

- We don't have labels or a deterministic measure of how alike images are

Prompt: A Robot on Mars sipping on matcha in a sunset, caricature style

We classify generated images using a neural network!

Evaluation of Diffusion Models

- Frechet Inception Distance (FID)
  - Calculates the distance between distributions of images with known labels and generated images in the feature space from **InceptionNet**
  - Indication of the quality and diversity of generation
  - **Lower** FID values indicate the generated distribution is close to that of real data – a good thing!

$$\mathrm{FID} = ||\mu_r - \mu_g||^2 + \mathrm{Tr}(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2})$$

Evaluation of Diffusion Models

- Inception Score (IS)
    - Leverages the classification output of an Inception Network
    - The score takes into account classification confidence and diversity
    - **Higher** IS values indicate the generated distribution is of high quality

$$\text{IS} = \exp\left(\mathbb{E}_x\left[D_{\text{KL}}(p(y|x)||p(y))\right]\right)$$

Exploring advanced techniques

Recap: We now know the basic settings to train a diffusion model

Task: You're now encouraged to explore different directions to build upon the quality of your diffusion model!

👉 Use a learnt variance for your DDPM

👉 Explore different noise schedules and log their effects

👉 Play with the model architecture: Try using Diffusion Transformers!

Kaggle Competition

Submit Inception-v3 feature statistics of your generated images to Kaggle InClass competition

Submission format: CSV file, not raw images

👉 Generate 5,000 images (50 per class with CFG, or 5,000 unconditional)

👉 Extract 2048-dim pool3 features via Inception-v3

👉 Use generate_submission.py to create CSV from mean and covariance

Kaggle evaluates FID between your stats and a hidden test reference. Lower FID = better.

Diffusion Techniques Competition

Goal: Systematically benchmark different diffusion training strategies on a shared dataset and evaluation protocol. Top results and findings may be compiled into a competition-style paper.

What this means for you:

👉 Document your technique, hyperparameters, and results clearly in your report

👉 Report FID/IS for each ablation so results are comparable across students

👉 Strong submissions may be invited to co-author the compiled paper

Official release Sun Feb 22 @ 6pm EDT

Here's some resources we organized btw

https://docs.google.com/document/d/1N3TtMNlUt9EoQbmb53_qs5N5Br6F8mj9F8nyzjAbcM8/

Questions?